



**CANCER EPIGENETICS COURSE**

# **INTRODUCTION TO R PROGRAMMING LANGUAGE**

Presenter: Tran Duc Phuc

December 7th 2025

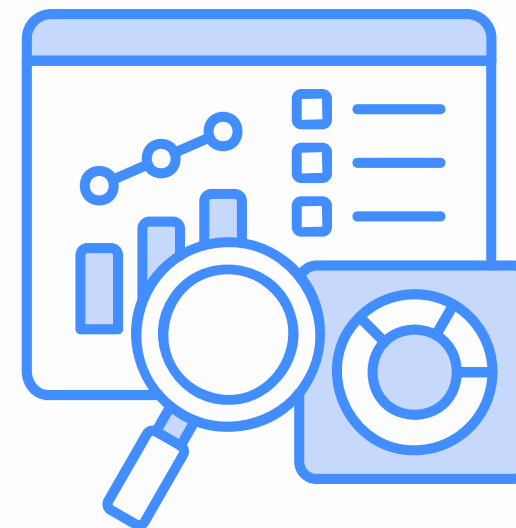
# INTRODUCTION

# R Programming Language

- Open-source programming language
- Main functions:
  - Statistical analysis
  - Data visualization



**Main function**



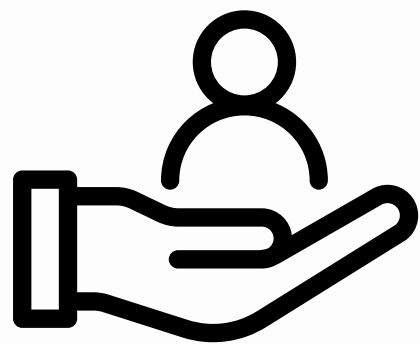
Statistical analysis



Visualization

# Advantages

## Customizable



- Highly customize
- Multiple packages

## Free open-source



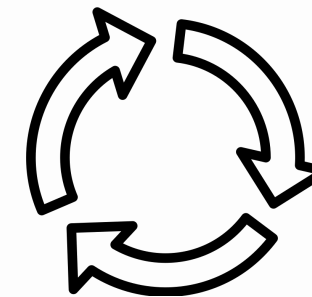
- Completely free

## Active community support



- Multiple forums, community
- Help debugging

## Reproducible



- Ex: using 1 script for many studies

## Simple



- Simpler than the majority CLI software

# Installation and Activation

**Installation (for Linux):** `sudo apt install r-base`

**Activation in terminal**

```
phuc@phuc-TDP:~$ R

R version 4.5.2 (2025-10-31) -- "[Not] Part in a Rumble"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> 
```

<https://www.r-project.org>



# R Studio



- An open source integrated development environment (IDE) for R programming language
- Main functions: provide a better working space for R programming
- Download from R Studio website:  
<https://posit.co/download/rstudio-desktop/>
- `sudo gdebi <file_name>`

| OS                  | Download                                           | Size      | SHA-256                  |
|---------------------|----------------------------------------------------|-----------|--------------------------|
| Windows 10/11       | <a href="#">RSTUDIO-2025.05.1-513.EXE ±</a>        | 281.24 MB | <a href="#">3A553330</a> |
| macOS 13+           | <a href="#">RSTUDIO-2025.05.1-513.DMG ±</a>        | 607.30 MB | <a href="#">76E1538B</a> |
| Ubuntu 22/Debian 12 | <a href="#">RSTUDIO-2025.05.1-513-AMD64.DEB ±</a>  | 209.78 MB | <a href="#">89A68B37</a> |
| Ubuntu 24           | <a href="#">RSTUDIO-2025.05.1-513-AMD64.DEB ±</a>  | 209.78 MB | <a href="#">89A68B37</a> |
| Fedora 41           | <a href="#">RSTUDIO-2025.05.1-513-X86_64.RPM ±</a> | 224.98 MB | <a href="#">6A97DF24</a> |

```
phuc@phuc-TDP: ~  
phuc@phuc-TDP:~$ sudo gdebi rstudio-2025.05.1-513-amd64.deb
```

File Edit Code View Plots Session Build Debug Profile Tools Help

DESeq2\_mtest.R

Untitled1

R data sets

info

CO2

R data sets

info

Run

Source

1

Environment

History

Connections

Tutorial

Import Dataset

2.11 GiB

List

R

Global Environment

Data

•annot\_info

37 obs. of 2 variables

•avg\_annot\_in

8 obs. of 1 variable

•brain

91456 obs. of 9 variables

•brain\_specif

100 obs. of 8 variables

•coldata

38 obs. of 3 variables

•coldata\_filt

37 obs. of 3 variables

•count\_matrix

Large matrix (4768050 elements, 49\_

•count\_matrix

Large matrix (4642575 elements, 48\_

•dds

Large DESeqDataSet (120023 element\_

Files

Plots

Packages

Help

Viewer

Presentation

New Folder

New File

Delete

Rename

More

Home

|  | Name                                | Size     | Modified              |
|--|-------------------------------------|----------|-----------------------|
|  | .RData                              | 299.1 MB | Dec 7, 2025, 4:30 PM  |
|  | .Rhistory                           | 17.8 KB  | Dec 7, 2025, 4:53 PM  |
|  | apps                                |          |                       |
|  | Desktop                             |          |                       |
|  | Documents                           |          |                       |
|  | Downloads                           |          |                       |
|  | miniconda3                          |          |                       |
|  | Music                               |          |                       |
|  | plam.txt                            | 565 KB   | Dec 5, 2025, 1:28 PM  |
|  | phuctran_id_rna.pub                 | 749 B    | Nov 25, 2025, 8:59 AM |
|  | Pictures                            |          |                       |
|  | Public                              |          |                       |
|  | R                                   |          |                       |
|  | snap                                |          |                       |
|  | statistics.sh                       | 471 B    | Dec 2, 2025, 1:18 PM  |
|  | tech                                | 494 B    | Dec 4, 2025, 9:07 AM  |
|  | Templates                           |          |                       |
|  | Thesis                              |          |                       |
|  | Venn Diagram of Annotated Genes.png | 359 KB   | Dec 6, 2025, 4:11 AM  |
|  | Videos                              |          |                       |
|  | Zotero                              |          |                       |

1:1

(Top Level)

R Script

Console

Terminal

Background jobs

R 4.3.3

~/R

Natural language support but running in an English locale

R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

>

# **VARIABLES IN R**



# Variables

- Variables store data with named locations
- Assigning variable syntax: `x <- 10; y <- "abc";...`
- Variable name contains: *letters, digits, underscore*
- Case sensitive
- **Must not**: start with digits, underscore; contains special characters

## Valid variable name

total

sum

a\_b\_c

var

Var

.count

## Invalid variable name

tot@l

s u m

5\_ab

.0var

TRUE

\_count

## Example

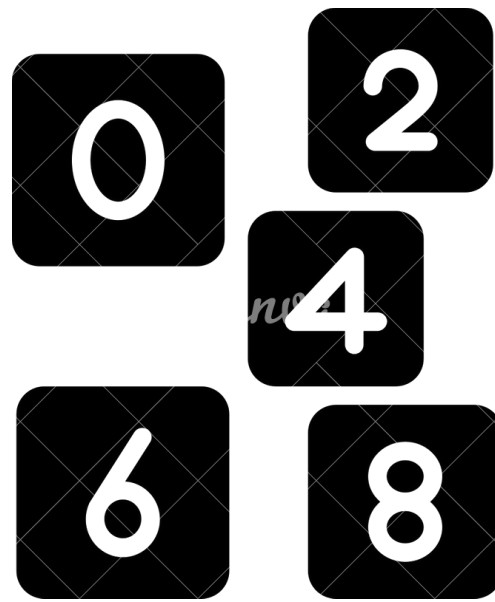
```
> x <- 10  
> .y <- 20  
> z <- x+.y  
> print (z)  
[1] 30
```

```
> tot@l <- 30
```

```
Error: object 'tot' not found
```

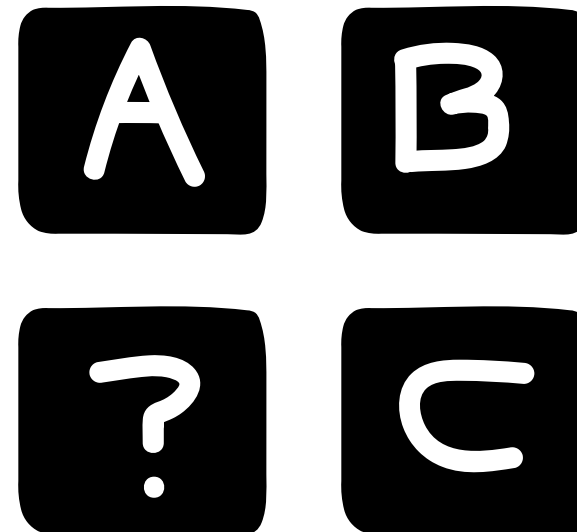
# Basic data types in R

## 1 Numeric



- All real number
- Including whole and decimal numbers
- Ex: myage <- 22

## 2 Character



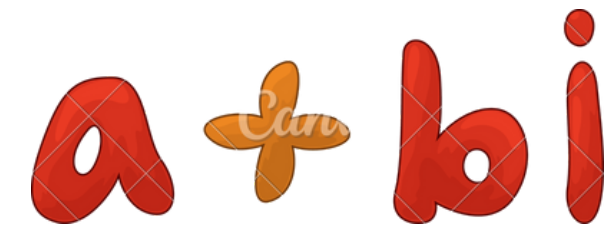
- String values encase in " "
- Ex: myname <- "Phuc"

## 3 Logical



- TRUE (T) or FALSE (F)
- Ex: graduate <- FALSE

## 4 Complex



- Complex nubmer
- Ex: 3 + i;...

- Classing variable's type: *class* (<variable>)

# LOGICAL OPERATORS

# Basic math

| Operator       | Description               | Example     | Result |
|----------------|---------------------------|-------------|--------|
| <b>+</b>       | Addition                  | 5 + 3       | 8      |
| <b>-</b>       | Subtraction               | 12 - 6      | 6      |
| <b>*</b>       | Multiplication            | 6 * 7       | 42     |
| <b>/</b>       | Division                  | 12 / 4      | 3      |
| <b>^ or **</b> | Exponentiation<br>(Power) | 2^3 or 2**3 | 8      |
| <b>%%</b>      | Modulo<br>(Remainder)     | 10 %% 3     | 1      |

# Comparison operators

| Operator     | Description              | Example |
|--------------|--------------------------|---------|
| <b>==</b>    | Equal to                 | x == 10 |
| <b>!=</b>    | Not equal to             | x != 10 |
| <b>&gt;</b>  | Greater than             | x > 10  |
| <b>&lt;</b>  | Less than                | x < 10  |
| <b>&gt;=</b> | Greater than or equal to | x >= 10 |
| <b>&lt;=</b> | Less than or equal to    | x <= 10 |

# Logical operators

| Operator     | Description                                | Example           | Result |
|--------------|--------------------------------------------|-------------------|--------|
| <b>&amp;</b> | And. TRUE if every conditions are TRUE     | 10 > 20 & 10 < 20 | FALSE  |
| <b> </b>     | Or. TRUE if at least one condition is TRUE | 10 > 20   10 < 20 | TRUE   |
| <b>!</b>     | Not. Opposite logical value                | ! (10 < 20)       | FALSE  |



## If statement

- Syntax:

if (<condition>) *#Assign a condition*

{<action1>} *#Condition is met, execute action1*

else {action2} *#Condition is not met, execute action1*

```
> myage <- 22
> if (myage > 18) {
+   print("I am an adult")
+ } else {print("I am a junior")}
+ }
[1] "I am an adult"
```

```
> myage <- 14
> if (myage > 18) {
+   print("I am an adult")
+ } else {print("I am a junior")}
+ }
[1] "I am a junior"
```

## Looping with “for”

- Looping action until end of list
- Syntax:

```
for (<variable> in <list>)  
{action}
```

*# List could be a list of string variables,  
list of numbers*

```
> for (i in 0:10) {  
+   print (i)  
+ }  
[1] 0  
[1] 1  
[1] 2  
[1] 3  
[1] 4  
[1] 5  
[1] 6  
[1] 7  
[1] 8  
[1] 9  
[1] 10  
>
```

## Looping with “while”

- Looping action until the assigned condition is broken
- Syntax:

```
while (<condition>)  
{action}
```

*# List could be a list of string variables,  
list of numbers*

```
> k <- 0  
> while (k != 10) {  
+   print (k)  
+   k=k+1}  
[1] 0  
[1] 1  
[1] 2  
[1] 3  
[1] 4  
[1] 5  
[1] 6  
[1] 7  
[1] 8  
[1] 9
```



# Data import and export

**Import data:** `read.csv(<path>)/`

`read.delim(<path>)/ read.table(<path>)`

*#path could be a download link or a directory*

```
> read <- read.csv("/home/phuc/Downloads/clinical.csv")
> print(read)
```

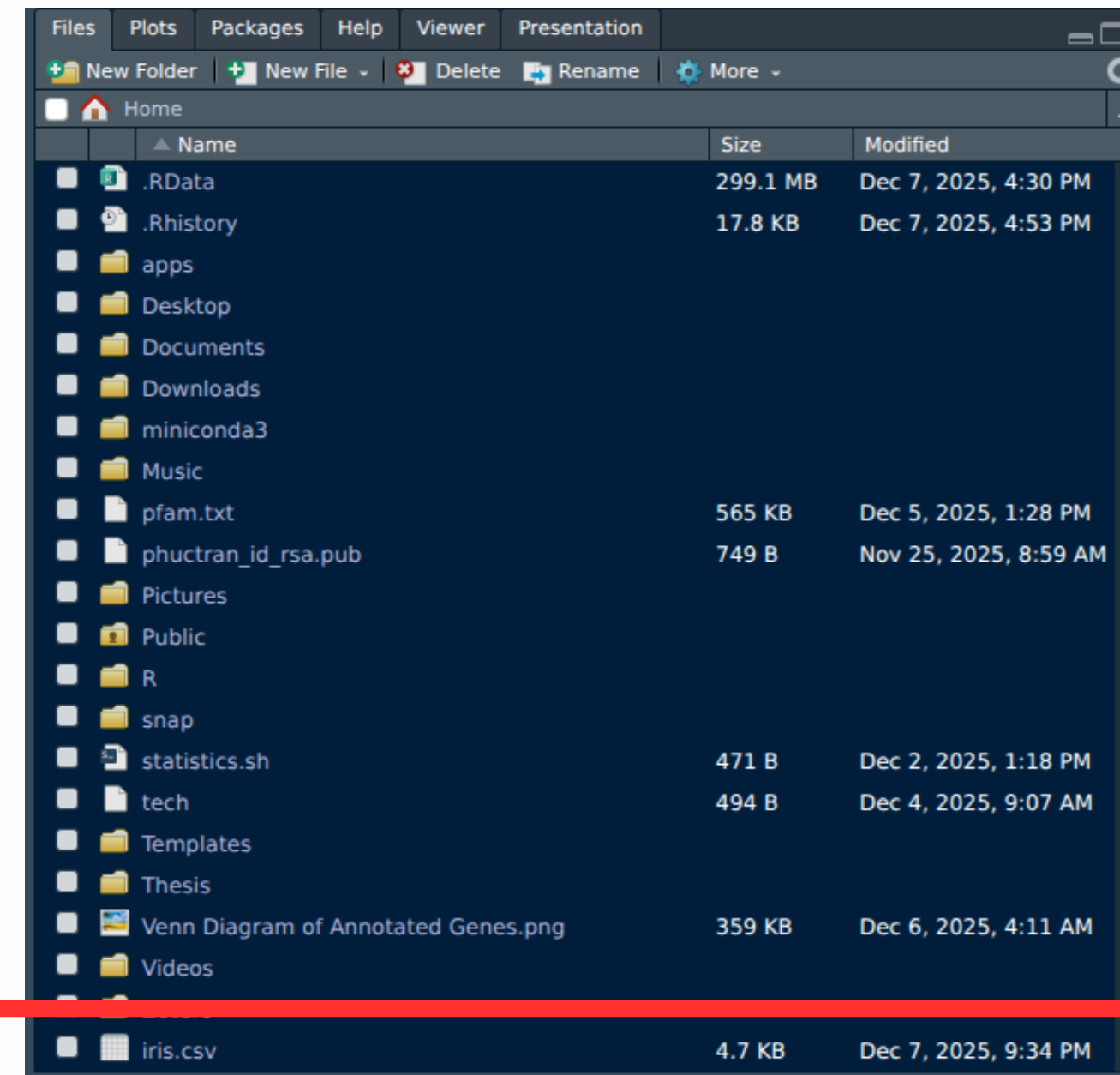
|    | CaseNo | Gender | Age | Survival.Months. |
|----|--------|--------|-----|------------------|
| 1  | 1      | F      | 71  | 1                |
| 2  | 2      | M      | 70  | 13               |
| 3  | 3      | F      | 73  | 8                |
| 4  | 4      | M      | 64  | 14               |
| 5  | 7      | F      | 74  | 4                |
| 6  | 8      | F      | 84  | 0                |
| 7  | 9      | F      | 86  | 1                |
| 8  | 12     | M      | 49  | 15               |
| 9  | 13     | M      | 76  | 1                |
| 10 | 14     | F      | 63  | 7                |
| 11 | N1     | F      | 62  | N/A              |
| 12 | N2     | M      | 64  | N/A              |
| 13 | N3     | F      | 40  | N/A              |
| 14 | N4     | F      | 56  | N/A              |

**Export data:** `write.csv(<data>,<dir>)/`

`write.delim(<data>,<dir>)/`

`write.table(<data>,<dir>)`

```
> write.csv (iris, "/home/phuctran/iris.csv")
```



| Name                                | Size     | Modified              |
|-------------------------------------|----------|-----------------------|
| .RData                              | 299.1 MB | Dec 7, 2025, 4:30 PM  |
| .Rhistory                           | 17.8 KB  | Dec 7, 2025, 4:53 PM  |
| apps                                |          |                       |
| Desktop                             |          |                       |
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| Videos                              |          |                       |
| iris.csv                            | 4.7 KB   | Dec 7, 2025, 9:34 PM  |

# Built-in dataset for practice

- All built-in datasets: data()
- View a dataset: View (<dataset>)

|                        |                                                                 |
|------------------------|-----------------------------------------------------------------|
| AirPassengers          | Monthly Airline Passenger Numbers 1949-1960                     |
| BJsales                | Sales Data with Leading Indicator                               |
| BJsales.lead (BJsales) | Sales Data with Leading Indicator                               |
| BOD                    | Biochemical Oxygen Demand                                       |
| CO2                    | Carbon Dioxide Uptake in Grass Plants                           |
| ChickWeight            | Weight versus age of chicks on different diets                  |
| DNase                  | Elisa assay of DNase                                            |
| EuStockMarkets         | Daily Closing Prices of Major European Stock Indices, 1991-1998 |
| Formaldehyde           | Determination of Formaldehyde                                   |
| HairEyeColor           | Hair and Eye Color of Statistics Students                       |
| Harman23.cor           | Harman Example 2.3                                              |
| Harman74.cor           | Harman Example 7.4                                              |
| Indometh               | Pharmacokinetics of Indomethacin                                |
| InsectSprays           | Effectiveness of Insect Sprays                                  |
| JohnsonJohnson         | Quarterly Earnings per Johnson & Johnson Share                  |
| LakeHuron              | Level of Lake Huron 1875-1972                                   |
| LifeCycleSavings       | Intercountry Life-Cycle Savings Data                            |
| Loblolly               | Growth of Loblolly pine trees                                   |
| Nile                   | Flow of the River Nile                                          |
| Orange                 | Growth of Orange Trees                                          |
| OrchardSprays          | Potency of Orchard Sprays                                       |
| PlantGrowth            | Results from an Experiment on Plant Growth                      |
| Puromycin              | Reaction Velocity of an Enzymatic Reaction                      |
| Seatbelts              | Road Casualties in Great Britain 1969-84                        |
| Theoph                 | Pharmacokinetics of Theophylline                                |
| Titanic                | Survival of passengers on the Titanic                           |
| ToothGrowth            | The Effect of Vitamin C on Tooth Growth in Guinea Pigs          |
| UCBAdmissions          | Student Admissions at UC Berkeley                               |
| UKDriverDeaths         | Road Casualties in Great Britain 1969-84                        |
| UKgas                  | UK Quarterly Gas Consumption                                    |

|    | ▲ | Sepal.Length | ↕ | Sepal.Width | ↕ | Petal.Length | ↕ | Petal.Width | ↕ | Species | ↕ |
|----|---|--------------|---|-------------|---|--------------|---|-------------|---|---------|---|
| 1  |   | 5.1          |   | 3.5         |   | 1.4          |   | 0.2         |   | setosa  |   |
| 2  |   | 4.9          |   | 3.0         |   | 1.4          |   | 0.2         |   | setosa  |   |
| 3  |   | 4.7          |   | 3.2         |   | 1.3          |   | 0.2         |   | setosa  |   |
| 4  |   | 4.6          |   | 3.1         |   | 1.5          |   | 0.2         |   | setosa  |   |
| 5  |   | 5.0          |   | 3.6         |   | 1.4          |   | 0.2         |   | setosa  |   |
| 6  |   | 5.4          |   | 3.9         |   | 1.7          |   | 0.4         |   | setosa  |   |
| 7  |   | 4.6          |   | 3.4         |   | 1.4          |   | 0.3         |   | setosa  |   |
| 8  |   | 5.0          |   | 3.4         |   | 1.5          |   | 0.2         |   | setosa  |   |
| 9  |   | 4.4          |   | 2.9         |   | 1.4          |   | 0.2         |   | setosa  |   |
| 10 |   | 4.9          |   | 3.1         |   | 1.5          |   | 0.1         |   | setosa  |   |
| 11 |   | 5.4          |   | 3.7         |   | 1.5          |   | 0.2         |   | setosa  |   |
| 12 |   | 4.8          |   | 3.4         |   | 1.6          |   | 0.2         |   | setosa  |   |
| 13 |   | 4.8          |   | 3.0         |   | 1.4          |   | 0.1         |   | setosa  |   |
| 14 |   | 4.3          |   | 3.0         |   | 1.1          |   | 0.1         |   | setosa  |   |
| 15 |   | 5.8          |   | 4.0         |   | 1.2          |   | 0.2         |   | setosa  |   |
| 16 |   | 5.7          |   | 4.4         |   | 1.5          |   | 0.4         |   | setosa  |   |
| 17 |   | 5.4          |   | 3.9         |   | 1.3          |   | 0.4         |   | setosa  |   |
| 18 |   | 5.1          |   | 3.5         |   | 1.4          |   | 0.3         |   | setosa  |   |
| 19 |   | 5.7          |   | 3.8         |   | 1.7          |   | 0.3         |   | setosa  |   |
| 20 |   | 5.1          |   | 3.8         |   | 1.5          |   | 0.3         |   | setosa  |   |

## Data inspection

| Syntax             | Function                                    |
|--------------------|---------------------------------------------|
| dim(<data>)        | Check for dimension of data (columns, rows) |
| colnames (<data>)  | Show every columns name                     |
| <data>\$<colnames> | Subset a column in dataset                  |

```
> dim (iris)
[1] 150  5
```

```
> colnames (iris)
[1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
```

```
> head (iris$Sepal.Length)
[1] 5.1 4.9 4.7 4.6 5.0 5.4
```

A top-down view of a desk with various items: a laptop in the upper right, a cup of coffee in the upper left, a pen in the center, a pair of glasses in the lower left, several paper clips on the left, and a large plant leaf in the bottom left corner. The background is a light, neutral color.

**THANK YOU  
FOR LISTENING**